Innovation-Based Sequential Detection of Radio Transients: Mathematical Foundations

KREMETART design notes

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Abstract

This note develops the statistical machinery underpinning the proposed transient detector, deliberately separated from the imaging and software-pipeline design. The detector treats each sky-pixel light curve as a Gaussian state-space model whose quiescent dynamics are an integrated Wiener process. A Kalman filter "whitens" the time series into a sequence of independent normalised innovations; a transient appears as a change in the distribution of those innovations, which we detect with sequential change-detection statistics (CUSUM and its generalised-likelihood-ratio extension). We define every quantity and acronym from first principles, give the discrete-time process matrices in closed form, derive the innovation likelihood and the marginal evidence, and characterise the operating point of the detector (detection delay, false-alarm rate, multiple testing). A multi-scale bank of models and its combination via Bayesian model averaging close the development.

1 Overview and the detection problem

We observe, at each sky pixel, a scalar light curve $\{y_k\}_{k=1}^{K}$ sampled at times t_k, where y_k is the (calibrated, source-subtracted) pixel flux in one frame. In the absence of a transient the light curve is a slowly varying "quiescent" signal plus measurement noise. A *transient* is a temporary additive flux excess a_k that switches on at an unknown time t_0 and persists for an unknown duration with unknown amplitude and shape.

The detector has three layers:

1. A *generative quiescent model* (Section 3): the light curve is an integrated Wiener process observed in noise. This encodes "what normal looks like" and provides a principled, recursive predictor.

2. A *whitening transform* (Section 4): the Kalman filter turns the correlated light curve into a sequence of independent, unit-variance *innovations* under the quiescent model. Detection reduces to testing whether those innovations remain zero-mean.

3. A *sequential test* (Section 8): CUSUM / GLR statistics accumulate the signed innovation evidence and raise an alarm when it crosses a calibrated threshold, while remaining primed for a change at any future time.

A bank of quiescent models at different temporal stiffnesses (Section 10) gives sensitivity across transient time-scales, in the manner of a multi-scale matched filter.

2 Notation and acronyms

Throughout, lower-case bold denotes column vectors, upper-case bold matrices, and $\mathcal{N}(\boldsymbol{m}, \boldsymbol{\Sigma})$ the Gaussian density with mean $\boldsymbol{m}$ and covariance $\boldsymbol{\Sigma}$. Table 1 collects the acronyms.

Acronym	Meaning
SDE	Stochastic differential equation
GP	Gaussian process
IWP	Integrated Wiener process (the quiescent prior)
KF	Kalman filter
LLR	Log-likelihood ratio
LR / NP	Likelihood ratio / Neyman–Pearson (lemma)
SPRT	Sequential probability ratio test (Wald)
CUSUM	Cumulative sum (change-detection) test (Page)
GLR	Generalised likelihood ratio (test)
NIS	Normalised innovation squared
NEES	Normalised estimation error squared
ARL	Average run length
ROC	Receiver operating characteristic
FDR	False discovery rate
BMA	Bayesian model averaging
RFI	Radio-frequency interference

Table 1: Acronyms used in this note.

3 The quiescent model: an integrated Wiener process

Continuous-time prior. A *stochastic differential equation* (SDE) describes a continuous-time random process through an ordinary differential equation driven by white noise. We model the quiescent flux $f(t)$ as the q-times *integrated Wiener process* (IWP): the q-th derivative of f is a Wiener process (Brownian motion). Collecting f and its first q derivatives into a state vector $\boldsymbol{x}(t) = \big(f(t),\, f^{(1)}(t), \ldots, f^{(q)}(t)\big)^{\mathsf{T}} \in \mathbb{R}^{q+1}$, the IWP is the linear SDE

$$\mathrm{d}\boldsymbol{x}(t) \;=\; \boldsymbol{F}\,\boldsymbol{x}(t)\,\mathrm{d}t \;+\; \boldsymbol{L}\,\mathrm{d}\beta(t), \qquad \boldsymbol{F} = \begin{pmatrix} 0 & 1 & & \\ & 0 & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{pmatrix}, \quad \boldsymbol{L} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}, \tag{1}$$

where $\beta(t)$ is a scalar Wiener process with diffusion (spectral density) σ^2. The scalar σ^2 is the *driving variance*: it sets how rapidly the top derivative wanders, hence how "stiff" or "flexible" the light curve is. The IWP is a *Gauss–Markov process*—Gaussian, and Markov in the state $\boldsymbol{x}$—which is exactly what makes recursive (Kalman) inference exact and $O(1)$ per sample.

Remark 1 (Relation to Matérn Gaussian processes and "length scale")**.** *The IWP is non-stationary: its variance grows without bound, so it has no stationary length scale. It is the natural smoothness prior of probabilistic numerics. A* stationary *alternative with an explicit length scale ℓ is the Matérn Gaussian process (GP) of half-integer smoothness $\nu = q + \frac{1}{2}$, which admits an SDE representation of the same order* $q{+}1$ *(the companion matrix $\boldsymbol{F}$ gains a stable characteristic polynomial set by ℓ).*

In the bank of Section 10 the "scales" are realised by varying σ^2 (and optionally the order q) for the IWP, or ℓ for the Matérn variant. We use "stiffness" for the qualitative knob: smaller σ^2 (or larger ℓ) = stiffer = slower to follow changes.

Exact discretisation. For a sampling interval $\Delta_k = t_k - t_{k-1}$, the SDE (1) integrates *exactly* to a discrete linear-Gaussian transition

$$\boldsymbol{x}_k = \boldsymbol{A}(\Delta_k)\,\boldsymbol{x}_{k-1} + \boldsymbol{w}_k, \qquad \boldsymbol{w}_k \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}(\Delta_k)), \tag{2}$$

with $\boldsymbol{A}(\Delta) = \exp(\boldsymbol{F}\Delta)$ and $\boldsymbol{Q}(\Delta) = \int_0^\Delta \exp(\boldsymbol{F}s)\,\boldsymbol{L}\sigma^2\boldsymbol{L}^\top \exp(\boldsymbol{F}s)^\top \mathrm{d}s$. Both are polynomials in Δ. Indexing rows/columns by derivative order $i, j \in \{0, \dots, q\}$,

$$A_{ij}(\Delta) = \frac{\Delta^{\,j-i}}{(j-i)!} \quad (j \geq i), \quad 0 \text{ otherwise}; \qquad Q_{ij}(\Delta) = \sigma^2\,\frac{\Delta^{\,2q-i-j+1}}{(q-i)!\,(q-j)!\,(2q-i-j+1)}. \tag{3}$$

The two cases used in practice are the constant-velocity (q=1) and constant-acceleration (q=2) models:

$$q{=}1: \ \boldsymbol{A} = \begin{pmatrix} 1 & \Delta \\ 0 & 1 \end{pmatrix}, \ \boldsymbol{Q} = \sigma^2 \begin{pmatrix} \Delta^3/3 & \Delta^2/2 \\ \Delta^2/2 & \Delta \end{pmatrix}; \quad q{=}2: \ \boldsymbol{A} = \begin{pmatrix} 1 & \Delta & \Delta^2/2 \\ 0 & 1 & \Delta \\ 0 & 0 & 1 \end{pmatrix}, \ \boldsymbol{Q} = \sigma^2 \begin{pmatrix} \frac{\Delta^5}{20} & \frac{\Delta^4}{8} & \frac{\Delta^3}{6} \\ \frac{\Delta^4}{8} & \frac{\Delta^3}{3} & \frac{\Delta^2}{2} \\ \frac{\Delta^3}{6} & \frac{\Delta^2}{2} & \Delta \end{pmatrix}.$$

For uniform sampling, $\boldsymbol{A}$ and $\boldsymbol{Q}$ are computed once per scale.

4 Observation model and the Kalman filter

We observe the flux (the zeroth derivative) in additive Gaussian noise:

$$y_k = \boldsymbol{H}\boldsymbol{x}_k + v_k, \qquad \boldsymbol{H} = (1, 0, \dots, 0), \qquad v_k \sim \mathcal{N}(0, R_k), \tag{4}$$

where R_k is the per-frame noise variance, taken from the imaging weights / thermal sensitivity (it may vary across pixels and frames). Together (2) and this equation form a *linear-Gaussian state-space model*, for which the *Kalman filter* (KF) gives the exact filtering distribution $p(\boldsymbol{x}_k \,|\, y_{1:k}) = \mathcal{N}(\boldsymbol{x}_{k|k}, \boldsymbol{P}_{k|k})$ recursively.

Recursion. With $\boldsymbol{x}_{k|k-1} = \boldsymbol{A}\boldsymbol{x}_{k-1|k-1}$ and $\boldsymbol{P}_{k|k-1} = \boldsymbol{A}\boldsymbol{P}_{k-1|k-1}\boldsymbol{A}^\top + \boldsymbol{Q}$ (prediction):

$$e_k = y_k - \boldsymbol{H}\boldsymbol{x}_{k|k-1} \qquad (\textit{innovation}\text{: one-step prediction error}), \tag{5}$$

$$S_k = \boldsymbol{H}\boldsymbol{P}_{k|k-1}\boldsymbol{H}^\top + R_k \qquad (\textit{innovation variance}), \tag{6}$$

$$\boldsymbol{K}_k = \boldsymbol{P}_{k|k-1}\boldsymbol{H}^\top S_k^{-1} \qquad (\textit{Kalman gain}), \tag{7}$$

$$\boldsymbol{x}_{k|k} = \boldsymbol{x}_{k|k-1} + \boldsymbol{K}_k e_k, \quad \boldsymbol{P}_{k|k} = (\boldsymbol{I} - \boldsymbol{K}_k\boldsymbol{H})\boldsymbol{P}_{k|k-1} \qquad (\textit{update}). \tag{8}$$

For numerical robustness across many parallel filters use the Joseph form $\boldsymbol{P}_{k|k} = (\boldsymbol{I} - \boldsymbol{K}_k\boldsymbol{H})\boldsymbol{P}_{k|k-1}(\boldsymbol{I} - \boldsymbol{K}_k\boldsymbol{H})^\top + \boldsymbol{K}_k R_k \boldsymbol{K}_k^\top$, or a square-root filter.

The whitening property. The decisive fact for detection is that, *under the model*, the innovation sequence is zero-mean and temporally white (Kailath's innovations approach):

$$\mathbb{E}[e_k] = 0, \qquad \mathbb{E}[e_k e_j] = S_k\, \delta_{kj}. \tag{9}$$

The KF therefore acts as a *whitening filter*: it removes all the predictable (quiescent) structure, leaving a white residual. Anything that is *not* explained by the quiescent model survives in the innovations. Detection becomes a test on the innovation sequence rather than on the raw, correlated light curve.

5 The Bayesian evidence (prediction-error decomposition)

Because the innovations are independent Gaussians, the marginal likelihood of the data under the model—the *Bayesian evidence*—factorises as the product of the one-step predictive densities (the "prediction-error decomposition"):

$$\log p(y_{1:K} \,|\, \theta) = \sum_{k=1}^{K} \log \mathcal{N}(y_k; \boldsymbol{H}\boldsymbol{x}_{k|k-1}, S_k) = -\tfrac{1}{2} \sum_{k=1}^{K} \Big[\log(2\pi S_k) + \tfrac{e_k^2}{S_k} \Big], \tag{10}$$

where $\theta = (q, \sigma^2, \dots)$ are the model hyperparameters. This is the evidence "that falls out of the Kalman filter." It serves two roles: (i) hyperparameter inference, by maximising or marginalising (10) over θ (the objective is differentiable, so gradient methods or a grid both work); and (ii) weighting the model bank in Section 10. Note that for the linear-Gaussian model (10) is *exact in closed form*; no particle/importance sampling is required to evaluate or marginalise it.

6 Innovation consistency: NIS and NEES

Define the *normalised innovation*

$$z_k \;=\; \frac{e_k}{\sqrt{S_k}} \;\sim\; \mathcal{N}(0,1) \quad \text{under the quiescent model,} \tag{11}$$

and the *normalised innovation squared* (NIS)

$$\epsilon_k \;=\; e_k^\top S_k^{-1} e_k \;=\; z_k^2 \;\sim\; \chi^2_{n_y}, \tag{12}$$

where $\chi^2_{n_y}$ is the chi-squared distribution with n_y degrees of freedom and $n_y = \dim(y_k) = 1$ here. (The state-space analogue using the true state error, $(\boldsymbol{x}_k - \boldsymbol{x}_{k|k})^\top \boldsymbol{P}_{k|k}^{-1} (\boldsymbol{x}_k - \boldsymbol{x}_{k|k}) \sim \chi^2_{n_x}$, is the *normalised estimation error squared*, NEES; it requires ground truth and is used in simulation to verify filter consistency.)

A time-averaged NIS over W samples, $\bar{\epsilon} = \frac{1}{W} \sum \epsilon_k$, has mean n_y and lies within $[\,\chi^2_{n_y W}(\frac{\alpha}{2}),\, \chi^2_{n_y W}(1-\frac{\alpha}{2})\,]/W$ with probability $1-\alpha$ under the model. This is a *consistency check*: before trusting any detection, confirm the quiescent innovations are white and unit-variance (NIS in bounds; flat autocorrelation of z_k). NIS is, however, a two-sided single-sample statistic—it discards the sign of the innovation and does not accumulate evidence—so it is a poor detector of faint or slow transients. The detector proper accumulates the *signed* innovations, as developed next.

7 The transient as a change in distribution

Write the pixel flux as quiescent plus transient, $f_k = b_k + a_k$, with $a_k = 0$ for $k < t_0$. The KF is built to track b_k. By linearity of the filter, an additive input a_k injects a *deterministic* mean into the innovation sequence:

$$\mu_k \;=\; \mathbb{E}[e_k \,|\, a_{1:k}] \;=\; a_k - \boldsymbol{H}\boldsymbol{A}\,\boldsymbol{x}^a_{k-1|k-1}, \tag{13}$$

where $\boldsymbol{x}^a$ is the state estimate the filter would form when driven by a alone. In words: the innovation sees the part of the transient that the filter has *not yet absorbed* into its state. The hypotheses are therefore

$$H_0 :\; z_k \sim \mathcal{N}(0,1) \quad \text{for all } k; \qquad H_1 :\; z_k \sim \mathcal{N}(\delta_k, 1) \text{ for } k \geq t_0, \quad \delta_k = \mu_k/\sqrt{S_k}. \tag{14}$$

Remark 2 (Why stiffness controls slow-transient detectability)**.** *For a step transient $a_k = A\,\mathbf{1}[k \geq t_0]$, equation* (13) *shows that $\mu_{t_0} = A$ and that μ_k then decays toward zero as the filter "catches up", with a time constant set by the closed-loop gain—hence by the stiffness. A* flexible *model (large σ^2) absorbs the step within a few samples, $\delta_k \to 0$ quickly, and a slow/faint transient becomes invisible: it is* explained away*. A* stiff *model (small σ^2) lags, keeping δ_k elevated over many samples, so the excess accumulates in the test statistic. Slow transients thus require both a stiff quiescent reference and a long integration horizon—two facets of the same requirement, and the motivation for the multi-scale bank. The cost of excessive stiffness is false alarms on benign slow drifts (gains, ionosphere, intrinsic variability).*

8 Sequential detection

Likelihood ratio and Neyman–Pearson (background). For deciding between two simple hypotheses, the Neyman–Pearson (NP) lemma states that the most powerful test at a given false-alarm probability rejects H_0 when the *likelihood ratio* (LR) exceeds a threshold. For Gaussian innovations, the per-sample *log-likelihood ratio* (LLR) for a shift of size δ is

$$\ell_k(\delta) = \log \frac{\mathcal{N}(z_k; \delta, 1)}{\mathcal{N}(z_k; 0, 1)} = \delta z_k - \tfrac{1}{2}\delta^2. \tag{15}$$

SPRT (Wald). The *sequential probability ratio test* accumulates $\Lambda_k = \sum_{i \leq k} \ell_i$ and stops as soon as $\Lambda_k \geq \log \frac{1-\beta}{\alpha}$ (declare H_1) or $\Lambda_k \leq \log \frac{\beta}{1-\alpha}$ (declare H_0), where α is the type-I error probability (false alarm) and β the type-II error probability (miss). Among tests with these error rates, the SPRT minimises the expected number of samples. Its limitation for our use is that it assumes a *known* change time and a single resolved decision; transient monitoring needs continuous surveillance with an *unknown* onset.

CUSUM (Page's test). The *cumulative sum* test handles an unknown onset by resetting the accumulation whenever the evidence turns non-positive:

$$g_k = \max\bigl(0,\; g_{k-1} + \ell_k(\delta)\bigr), \qquad g_0 = 0, \qquad \text{alarm when } g_k \geq h. \tag{16}$$

Substituting (15) gives the equivalent one-sided "tabular" form for a positive (brightening) shift,

$$g_k = \max\bigl(0,\; g_{k-1} + z_k - \kappa\bigr), \qquad \kappa = \delta/2, \tag{17}$$

where the *reference value* κ is half the smallest standardised shift one wishes to detect promptly. The $\max(0, \cdot)$ keeps the statistic primed for a change beginning at any future sample; CUSUM is equivalent to an SPRT restarted at every candidate change time. It is minimax optimal in Lorden's sense: it minimises the worst-case expected detection delay subject to a lower bound on the mean time between false alarms. A one-sided statistic (positive shifts only) is appropriate for brightening transients and is roughly twice as sensitive as the two-sided version.

GLR (generalised likelihood ratio). CUSUM fixes the shift size δ. When the amplitude is unknown, the *generalised likelihood ratio* test maximises the LLR over both the unknown onset t_0 and the unknown amplitude. For a Gaussian mean shift the inner maximisation is the maximum-likelihood estimate of δ over a candidate segment, with closed form

$$G_k = \max_{1\leq L\leq N} \frac{1}{2L}\left(\sum_{i=k-L+1}^{k} z_i\right)^2, \qquad \text{alarm when } G_k \geq h, \tag{18}$$

where $L = k - t_0 + 1$ is the (unknown) elapsed duration and N is the maximum duration searched. A short L with large amplitude (a fast bright transient) and a long L with small amplitude (a slow faint transient) are both admissible, so the GLR adapts its integration length automatically. The maximising L estimates the duration and $\frac{1}{L}\sum z_i$ the standardised amplitude—useful science by-products. The search window N is precisely where a finite window legitimately enters the design: **it is the longest transient the detector can optimally integrate**, not a batching device. Lengthening N improves slow-transient sensitivity at the cost of computation and false-alarm exposure.

Matched-filter connection. If a transient temporal *shape* $\boldsymbol{s}$ (e.g. an exponential flare or a dispersed pulse) is known up to amplitude, the optimal statistic is the matched filter $(\boldsymbol{s}^\top \boldsymbol{z})^2/(\boldsymbol{s}^\top \boldsymbol{s})$, which is χ^2_1 under H_0. The GLR step test (18) is the special case $\boldsymbol{s} =$ boxcar; replacing the boxcar with a template bank generalises the detector to specific light-curve morphologies.

9 Operating characteristics and calibration

Average run length. The operating point is summarised by two *average run lengths* (ARLs): $\mathrm{ARL}_0 = \mathbb{E}[\text{samples to alarm} \mid H_0]$ (mean time between false alarms; to be made large) and $\mathrm{ARL}_1 = \mathbb{E}[\text{detection delay} \mid H_1]$ (to be made small). The threshold h trades them off; ARL_0 grows essentially exponentially in h. Closed-form approximations exist (Siegmund), but they assume exactly white Gaussian innovations and are best used only to seed an empirical calibration.

Receiver operating characteristic. The *receiver operating characteristic* (ROC) plots detection probability against false-alarm rate as h varies. We obtain it by *injection*: add synthetic transients of known amplitude, duration and morphology into real (source-subtracted) data and measure detections versus false alarms. This is also the end-to-end test against an independent imager (e.g. DiSkO).

Multiple testing. With P pixels each running a sequential test, the per-pixel threshold must be raised to control the *aggregate* false-alarm rate. For a target of one false alarm per T samples across the map, set the per-pixel $\mathrm{ARL}_0 \approx PT$ (a Bonferroni-type correction). When many candidate detections are expected, controlling the *false discovery rate* (FDR)—the expected fraction of declared detections that are spurious—via the Benjamini–Hochberg procedure is less conservative than controlling the family-wise error. Because the per-scale statistics (Section 10) are correlated, the multiplicity correction is best fixed by empirical calibration on quiescent residuals rather than analytic union bounds.

10 The multi-scale model bank and its combination

Run M quiescent models $\{\theta^{(m)}\}_{m=1}^{M}$ spanning a range of stiffnesses (values of σ^2, optionally orders q). Each KF emits its own innovation sequence $z_k^{(m)}$, evidence increment, and detection statistic $G_k^{(m)}$. Two combinations are useful:

Bayesian model averaging (BMA) for the quiescent description. Treating the bank as a discrete hyperparameter prior, the posterior model weights update recursively from the evidence increments (10):

$$w_k^{(m)} \propto w_{k-1}^{(m)} \mathcal{N}\big(y_k;\, \boldsymbol{H}\boldsymbol{x}_{k|k-1}^{(m)},\, S_k^{(m)}\big), \qquad \sum_m w_k^{(m)} = 1. \tag{19}$$

A forgetting factor (raising $w_{k-1}^{(m)}$ to a power < 1 before re-normalising) keeps the weights responsive to slow non-stationarity. *Critically*, this adaptation must run on a slower / more robust time-scale than detection: otherwise a transient is absorbed by up-weighting a flexible model, hiding the very signal of interest. In practice, freeze or heavily damp the weights during a candidate event.

Detection across scales. For declaring a transient, combine the per-scale detector outputs by the scale-wise maximum, $G_k = \max_m G_k^{(m)}$, with the threshold raised to absorb the M-fold multiplicity (again calibrated empirically). The argmax over scales reports the best-matched transient time-scale. This realises a temporal multi-scale matched filter: each scale is most sensitive to transients whose duration is comparable to, or longer than, that model's tracking time.

11 Practical considerations

- **Initialisation.** Start each filter from a diffuse prior ($\boldsymbol{P}_{0|0}$ large); the first few S_k are inflated, so suppress detections during a short warm-up.
- **Numerics.** Use Joseph-form or square-root covariance updates; with thousands of independent pixel×scale filters of state dimension $q+1 \le 3$, the per-frame cost is dominated by small fixed-size linear algebra and vectorises cleanly.
- **Robustness to RFI.** Heavy-tailed outliers violate the Gaussian observation model and inflate false alarms. A Student-t observation model or a simple innovation gate ($|z_k|$ clipped, with the sample down-weighted) restores robustness; the former makes the model non-Gaussian and is the one place a Rao–Blackwellised particle treatment (particles over an outlier/changepoint indicator, KF marginalising the continuous state) would be justified.
- **Validation order.** (i) Confirm filter consistency (NIS/whiteness) on quiescent data; (ii) calibrate h for the target false-alarm rate on source-subtracted residuals; (iii) build ROC curves by injection; (iv) cross-check the recovered flux against an independent imager.

12 Summary of the heuristic

For each pixel and each scale m: run the IWP Kalman filter (2), form the normalised innovation $z_k^{(m)}$, and accumulate the one-sided GLR statistic (18) over a window N matched to the slowest transient of interest. Declare a transient when $\max_m G_k^{(m)}$ exceeds a threshold calibrated for the desired map-wide false-alarm (or false-discovery) rate, reporting the onset, duration and amplitude from the maximising segment and the time-scale from the maximising model. Quiescent-model adaptation (BMA weights, σ^2) runs on a slow, damped time-scale so that transients are detected rather than absorbed.

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